Unit 1: Java Servlets & Request Handling

Questions

1. What is the primary difference between Core Java and Advanced Java?
2. What is a Java Servlet?
3. Name two common web or application servers used to run Servlet applications.
4. Differentiate between a static web page and a dynamic web page.
5. Which two packages contain the core classes and interfaces for the Servlet API?
6. Which class is typically extended to create an HTTP-based servlet?
7. List the main lifecycle stages of a Servlet in order.
8. How many times is the init() method called during a Servlet's lifecycle?
9. What is the main purpose of the destroy() method in a Servlet?
10. Which method handles client requests and delegates them based on request types (GET, POST)?
11. What is the role of the HttpServletRequest object?
12. Which method of HttpServletRequest is used to read data submitted from an HTML form field?
13. What is the role of the HttpServletResponse object?
14. Name two key differences between HTTP GET and POST methods.
15. What is Session Management and why is it necessary?
16. Name four session management techniques in Java Servlets.
17. What is a Cookie?
18. How does URL Rewriting work when browser cookies are disabled?
19. What is the difference between forward() and include() in RequestDispatcher?
20. What is a Filter in Java Servlets?

Unit 2: JavaServer Pages (JSP) & Architecture

Questions

1. What is JSP?
2. What happens to a JSP file when a browser requests it for the first time?

3. What are the main lifecycle stages of a JSP page?
4. Which JSP lifecycle method is executed for every single user request?
5. What is a JSP Directive?
6. Name the three main types of JSP directives.
7. Which attribute of the Page Directive is used to import Java packages into a JSP page?
8. What is the purpose of errorPage and isErrorPage attributes in JSP?
9. What is the Include Directive used for?
10. What is a JSP Scriptlet and what syntax does it use?
11. What is a JSP Declaration Tag and what syntax does it use?
12. What is a JSP Expression Tag and what syntax does it use?
13. Why should you avoid putting a semicolon (;) at the end of a JSP Expression tag?
14. Into which servlet method is code written inside a scriptlet (<% %>) automatically translated?
15. What does the MVC architecture stand for?
16. In a Web application using Servlets and JSP, which component acts as the Controller?
17. In MVC, which component acts as the View?
18. In MVC, what is the responsibility of the Model?
19. What is the main advantage of using the MVC design pattern in web development?
20. What is the difference between Scriptlets (<% %>) and Expression Tags (<%= %>) regarding output?

Unit 3: JDBC, EJB, JavaMail & WebSockets

Questions

1. What does JDBC stand for and what is its purpose?
2. Which type of JDBC driver is known as the "Thin Driver"?
3. List the primary steps required to execute a database query using JDBC.
4. What is the difference between Statement and PreparedStatement in JDBC?
5. Which JDBC interface stores data retrieved from a SQL SELECT query?
6. What does EJB stand for?
7. What are the main types of Enterprise Java Beans (EJB)?

8. What is the difference between a Stateful and a Stateless Session Bean?
9. What component manages EJB life cycle, transactions, and security automatically?
10. What is a JavaBean?
11. What is the purpose of Getter and Setter methods in a JavaBean?
12. What is the main protocol used by JavaMail API to send emails?
13. Which JavaMail class stores the email server configuration settings?
14. Name two key JAR files historically required to run JavaMail applications.
15. What is the main difference between traditional HTTP protocol and WebSocket protocol?
16. What is the initial handshake step in WebSocket communication?
17. Name the fundamental WebSocket lifecycle events.
18. Which annotation turns a Java class into a server endpoint for WebSockets?
19. What is JSON and why is it widely used?
20. What are two common Java libraries used for JSON processing?

Unit 4: Spring Boot & REST APIs

Questions

1. What is Spring Boot?
2. Name two key problems solved by Spring Boot compared to standard Spring applications.
3. What is an Embedded Server in Spring Boot?
4. Which annotation is used on the main class of a Spring Boot application?
5. What three annotations are combined inside @SpringBootApplication?
6. What is the purpose of Spring Boot Starter dependencies?
7. Which Starter dependency is used to build web applications and REST APIs?
8. What is Spring Boot Auto-Configuration?
9. What does IoC (Inversion of Control) mean in Spring?
10. What is a RESTful API?
11. What is the function of the @RestController annotation?
12. Name the primary HTTP methods used for CRUD operations in REST APIs.
13. Which Spring Boot annotation maps HTTP GET requests to a controller method?

14. What is the purpose of the @RequestBody annotation in a REST Controller?
15. What is the purpose of the @PathVariable annotation?
16. What role does @Service play in a Spring Boot application?
17. What role does @Repository play in a Spring Boot application?
18. What configuration file is used to specify properties like database credentials or server port in Spring Boot?
19. What tool can developers use to test REST APIs without developing a frontend UI?
20. How does Spring Boot JPA manage database tables automatically?